



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiiit@gmail.com, website: www.iiitsonapat.ac.in

Mid Sem-II

Branch: IT

M.-15 marks

Subject: COMMUNICATION SKILLS (HUL 204)

Time- 60 min.

Semester II 2023-27 Batch

Attempt All. (3 X 5 = 15)

Roll no. 12312021

1. What do you understand by the Thesis statement and the topic sentence? With the help of suitable examples, explain the different types of Thesis statements. Also, comment on the significance of the topic sentence and the thesis statement in writing.
2. Write a note on different types of writings. Discuss at least four different types of writings and explain the difference in structure and purpose of each type with the help of suitable examples.
3. Write a note on the differences between American English and British English with the help of differences in pronunciation and articulation in speech. Illustrate with the help of suitable examples.



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Mid Sem-II

Subject: Digital Electronics (ECL-202)

Roll no. /23/2021

Branch: IT

M.M.-15 marks

Time- 60 mint.

Note: All questions are compulsory.

Q-1	Design a combinational circuit using an appropriate decoder for the given Boolean expression: $F(x, y, z) = \bar{x} \cdot \bar{y} + x \cdot y + \bar{y} \cdot z$	(3 marks)
Q-2	Implement $F(A, B, C, D) = \sum m(0, 1, 5, 6, 8, 9, 12, 13)$ using multiplexer with n-1 select lines. Implement the table for circuit design also.	(3 marks)
Q-3	Describe the steps involved in comparing two 4-bit numbers $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$. Determine the required number of inputs and outputs for designing a 4-bit magnitude comparator, and provide the Boolean expression for the scenario where A is less than B.	(3 marks)
Q-4	A serial in/parallel out, 4-bit shift register initially contains all 1's. The data nibble 0101 is waiting to enter. What will be the output after three clock pulses?	(3 marks)
Q-5	Design a 3-bit synchronous up counter using D Flipflop, which count the sequence as (0, 2, 4, 6, 1, 3, 5, 7, 0, ...).	(3 marks)



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Mid Sem-II
Branch: CSE

Subject Name: Data Structure
2nd Semester

Subject code: CSE 203
M.M.-15 marks

Roll no. 12311034
Time- 60 min.

te: (1) Attempt all the questions.

(2) All questions carry equal marks (3*5=15)

Q-1	Describe the Queue Data Structure and its practical applications. Elaborate on the process of insertion and deletion in a linear queue. Furthermore, discuss the advantages of employing a circular queue compared to a linear queue.	(3 marks)
Q-2	Define and discuss the following terms: ii) Binary tree iii) Maximum and minimum Height of a binary tree with n nodes iv) Maximum and minimum number of nodes in a binary tree with height h	(3 marks)
Q-3	Insert the keys: 31, 21, 11, 16, 26, 24, 40, 36, and 44 into an initially empty Binary Search Tree (BST). Then, provide the Preorder, Inorder, and Postorder sequences for the resulting BST. Additionally, Write the pseudocode algorithms for Preorder, Inorder, and Postorder traversals.	(3 marks)
Q-4	Consider the following Binary Search Tree (BST) formed by inserting the keys in the order given: 20, 10, 30, 5, 15, 25, 35, 3, 7, 12, 18. Now, perform the following operations on this BST: a. Insert the key 22 into the BST. b. Delete the key 15 from the BST.	(3 marks)
Q-5	What is an AVL tree, and how does it offer advantages over a Binary Search Tree (BST)? Discuss the concept of balance factor (BF) and rotations. Then, illustrate these concepts by inserting the keys 20, 25, 29, 8, 3, 13, 27, 17, 16, 9, 1, 2, and 6 into an initially empty AVL tree.	(3 marks)



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Mid Sem-II
Branch: IT

Subject: Web Designing (ITC 205)
M.M.-15 marks

Roll no. [23]202
Time- 60 mint.

Note: All questions are compulsory.

Q-1	a. Define JavaScript and explain its role in web development. b. Differentiate between '=' and '==' operators in JavaScript. Provide examples to illustrate the difference. c. Write a simple JavaScript code snippet to display a "Hello, World!" message in a web browser.	(3 marks)
Q-2	a. Discuss the different data types available in JavaScript. Provide examples for each data type. b. Write a JavaScript function to calculate the area of a rectangle given its length and width as parameters.	(4 marks)
Q-3	a. What are JavaScript functions? Explain the difference between function declarations and function expressions. b. Describe the concept of scope in JavaScript. How does scope affect variable accessibility within functions?	(3 marks)
Q-4	a. Describe how you would select HTML elements using JavaScript. Provide examples for different selection methods. b. Write a JavaScript program that dynamically adds a new list item to an unordered list (' ') in response to a button click event.	(5 marks)



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Mid Sem-2

Subject: OOP C++

Roll no. 18312021

Branch: IT

M.M.-15 marks

Time- 60 mint.

Note: All questions are compulsory.

Q-1	Develop a program to model a university system with base classes like "Person" (for students, faculty, and staff) and derived classes like "Student", "Faculty", and "Staff". Implement appropriate member functions and demonstrate polymorphic behavior.	(3 marks)
Q-2	Develop a C++ program to encrypt a text file using a simple substitution cipher. Prompt the user to enter the name of the input file and the name of the output file for the encrypted text.	(3 marks)
Q-3	Discuss the limitations and drawbacks of multiple inheritance in C++ and strategies for addressing issues like the diamond problem and ambiguous base classes.	(3 marks)
Q-4	Discuss the concept of file buffering and its impact on I/O performance in C++. How can you control and optimize file buffering for different types of file operations?	(3 marks)
Q-5	Explain the concept of static polymorphism and dynamic polymorphism in C++. How are they achieved, and what are their advantages and disadvantages?	(3 marks)



Roll No:

1 2 3 1 2 0 2 1

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B.Tech II Sem II Mid Semester Examination, 22th April, 2024

Mid Sem-II

Subject: Mathematics-II

Course Code: MAL-201

Branch: IT

Maximum Marks. -15 marks

Time- 60 min.

Note: All questions are compulsory.

Q-1	Solve $(x^3 e^x - my^2)dx + mxydy = 0$.	(3marks)
Q-2	Solve $(1 - x^2) \frac{dy}{dx} + xy = xy^2$.	(3marks)
Q-3	Find the general solution of the differential equation: $(D^2 + n^2)y = \sec nx$.	(3marks)
Q-4	Solve $(x^2 D^2 - 3xD + 4)y = (1 + x)^2$.	(3marks)
Q-5	Solve the system of linear differential equations: $(3D^2 + 3D)x = 4t - 3$ and $(D - 1)x - D^2y = t^2$.	(3marks)